NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

INCH-POUND

MIL-PRF-83446/9D
6 July 1993
SUPERSEDING
MIL-C-83446/9C
12 May 1987

PERFORMANCE SPECIFICATION SHEET

COILS, RADIO FREQUENCY, CHIP, VARIABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-83446.

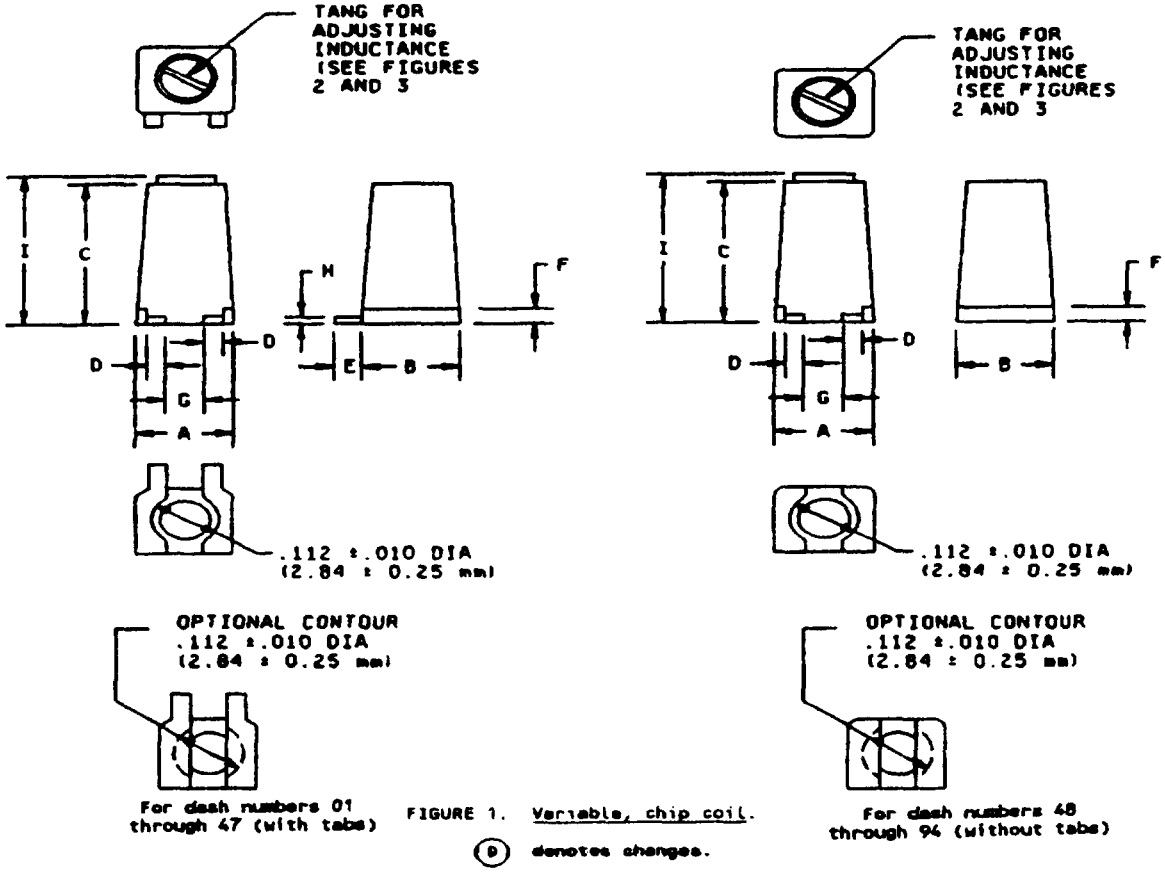


FIGURE 1. Variable, chip coil.
(D) denotes changes.

FIGURE 1. Variable, chip coil.

(D) denotes changes.

DISTRIBUTION STATEMENT A. Approved for public release, distribution is unlimited

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For dash numbers 01 through 47 (with tabs)		
Ltr	Inches	mm
A	.165 max	4.19 max
B	.130 max	3.30 max
C	.225 max	5.72 max
D	.030 ±.010	0.76 ±0.25
E	.040 +.015 -.010	1.02 +0.38 -0.25
F	.020 ±.005	0.51 ±0.13
G	.073 min	1.85 min
H	.005 ±.002	0.13 ±0.05
I	.250 max	6.35 max

For dash numbers 48 through 94 (without tabs)		
Ltr	Inches	mm
A	.165 max	4.19 max
B	.130 max	3.30 max
C	.225 max	5.72 max
D	.030 ±.010	0.76 ±0.25
F	.020 ±.005	0.51 ±0.13
G	.073 min	1.85 min
H	.005 ±.002	0.13 ±0.05
I	.250 max	6.35 max

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimension I is the maximum protrusion of the tang at minimum inductance.

FIGURE 1. Varible, chip coil - Continued.

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Weight: 0.5 gram maximum.

Operating temperature range: -55°C to +125°C.

Temperature rise (at 90°C): 35°C.

Maximum operating temperature: 125°C.

Altitude: 70,000 feet.

Dielectric withstanding voltage: Method 301 of MIL-STD-202, test voltage 700 volts rms.

Barometric pressure: Method 105, test condition C, MIL-STD-202 (70,000 feet), test voltage 200 volts rms.

Electrical characteristics (initial): See table I.

Electrical characteristics (final): See table II.

Supersession data: A new part numbering system with code for termination materials was incorporated with revision A, superseding MIL-I-83446/9, dated 1 March 1979.

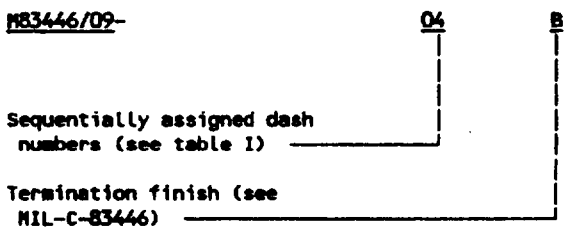
M83446/09-(dash number from table I) supersedes M83446/9-(dash number from table I).

Examples:

M83446/09-21C supersedes M83446/9-21.

M83446/09-03C supersedes M83446/9-3.

(D) Part or Identifying Number (PIN): The PIN shall be in the following form:



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TABLE I. Electrical characteristics (initial) and dash numbers.

Dash number	Inductance (µH)			Q Min at L Nom (D)	Q (typ) (at L nom) (D)	Test frequency (MHz) 1/	Self-resonant frequency (min) (MHz)	DC resistance at 25°C (max) (Ohms)	Current (max) (mA) 2/	Incremental current (max) (mA) 3/
	Min	Nom	Max							
01, 48	0.015	0.018	0.021	50	60	150	1200	0.03	750	- - -
02, 49	0.018	0.022	0.026	55	65	150	1100	0.03	750	- - -
03, 50	0.022	0.027	0.032	55	65	150	1000	0.03	750	- - -
04, 51	0.027	0.033	0.039	55	65	150	900	0.04	750	- - -
05, 52	0.033	0.039	0.047	55	65	150	750	0.04	750	- - -
06, 53	0.039	0.047	0.056	55	65	100	660	0.04	750	- - -
07, 54	0.047	0.056	0.068	55	65	100	610	0.05	700	- - -
08, 55	0.055	0.068	0.083	55	65	100	550	0.05	700	- - -
09, 56	0.067	0.082	0.099	55	65	100	520	0.06	645	- - -
10, 57	0.080	0.100	0.120	55	65	100	490	0.06	645	- - -
11, 58	0.096	0.120	0.144	50	60	50	460	0.07	600	- - -
12, 59	0.12	0.15	0.18	50	60	50	430	0.08	560	- - -
13, 60	0.14	0.18	0.22	50	60	50	385	0.09	525	- - -
14, 61	0.17	0.22	0.27	45	51	50	330	0.10	500	- - -
15, 62	0.21	0.27	0.33	45	51	50	300	0.15	400	- - -
16, 63	0.26	0.33	0.40	45	51	50	275	0.19	360	- - -
17, 64	0.32	0.39	0.47	40	50	50	240	0.25	300	- - -
18, 65	0.37	0.47	0.57	40	50	25	210	0.30	280	- - -
19, 66	0.46	0.56	0.68	40	50	25	190	0.40	250	- - -
20, 67	0.54	0.68	0.82	40	50	25	165	0.60	200	- - -
21, 68	0.68	0.82	1.00	40	50	25	155	0.70	180	- - -
22, 69	0.75	1.00	1.25	40	50	25	140	0.90	160	- - -
23, 70	0.90	1.20	1.50	35	44	25	125	1.1	150	- - -
24, 71	1.13	1.50	1.87	30	38	7.9	115	1.3	135	- - -
25, 72	1.35	1.80	2.25	30	38	7.9	105	1.7	120	- - -
26, 73	1.65	2.20	2.75	30	38	7.9	100	2.3	100	- - -
27, 74	2.02	2.70	3.37	25	35	7.9	85	3.0	90	- - -
28, 75	2.48	3.30	4.12	25	35	7.9	75	3.9	80	- - -

See footnotes at end of table.

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TABLE I. Electrical characteristics (initial) and dash numbers - Continued.

Dash number	Inductance (µH)			Q Min at L Nom	Q (typ) (at L nom)	Test frequency (MHz) 1/	Self-resonant frequency (min) (MHz)	DC resistance at 25°C (max) (Ohms)	Current (max) (mA) 2/	Incremental current (max) (mA) 3/
	Min	Nom	Max							
29, 76	2.92	3.9	4.87	25	33	7.9	47	3.9	80	- - -
30, 77	3.52	4.7	5.88	25	33	7.9	30	3.9	80	- - -
31, 78	4.2	5.6	7.0	22	30	7.9	24	4.0	75	- - -
32, 79	5.1	6.8	8.5	22	30	7.9	23	4.0	75	- - -
33, 80	6.1	8.2	10.2	22	30	7.9	22	4.5	72	- - -
34, 81	7.5	10.0	12.5	22	30	7.9	21	4.5	72	- - -
35, 82	9.0	12.0	15.0	22	30	7.9	17	5.0	70	- - -
36, 83	11.2	15.0	18.7	22	30	2.5	15	6.0	65	- - -
37, 84	13.5	18.0	22.5	22	30	2.5	14	8.0	55	- - -
38, 85	16.5	22.0	27.5	22	30	2.5	13	10.0	50	- - -
39, 86	20.2	27.0	31.3	22	30	2.5	11	11.0	45	- - -
40, 87	24.7	33.0	41.2	22	30	2.5	10	13.0	43	- - -
41, 88	29.2	39.0	48.8	22	30	2.5	9	14.0	42	- - -
42, 89	35.2	47.0	58.8	22	30	2.5	8	15.0	40	- - -
43, 90	42.0	56.0	70.0	22	30	2.5	7	19.0	36	- - -
44, 91	51.0	68.0	85.0	22	30	2.5	6	24.0	32	- - -
45, 92	62.0	82.0	102.0	25	33	2.5	5	31.0	28	5.6
46, 93	75.0	100.0	125.0	25	33	2.5	4	40.0	25	5.0
47, 94	90.0	120.0	150.0	25	33	2.5	3	50.0	22	4.4

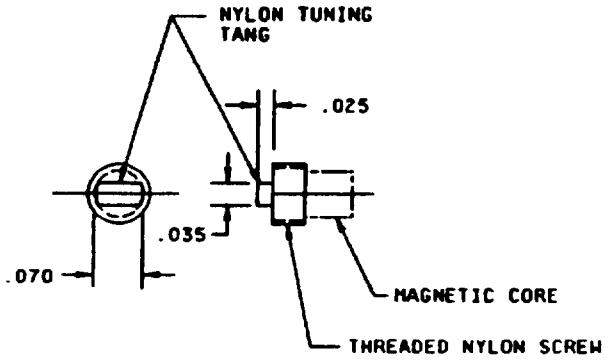
1/ Test frequency range 2.5 through 25 MHz: For electrical characteristics measurements, use the TF-260Q-3 test fixture, or equal (see figure 5). Fixture inductance (approximately 0.028 µH) and residual Q-meter inductance (approximately 0.01 µH) should be subtracted from indicated inductance.

Test frequency range 25 through 150 MHz: For electrical characteristics measurements, use the TF-250RX-3 test fixture, or equal (see figure 4). Fixture inductance (approximately 0.009 µH) should be subtracted from calculated inductance.

2/ Maximum current allowed is not to exceed the specified temperature rise.

3/ Applicable to ferrite core only.

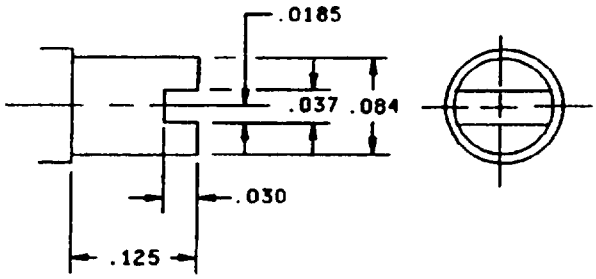
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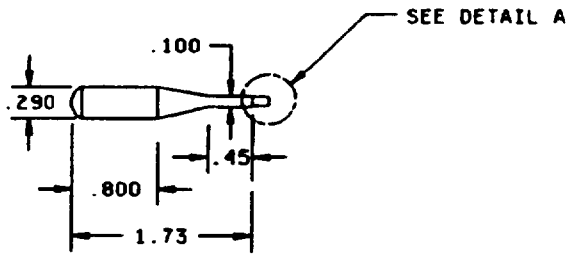
Inches	mm
.025	0.64
.035	0.89
.070	1.78

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 2. Core and adjustment screw assembly.



DETAIL A

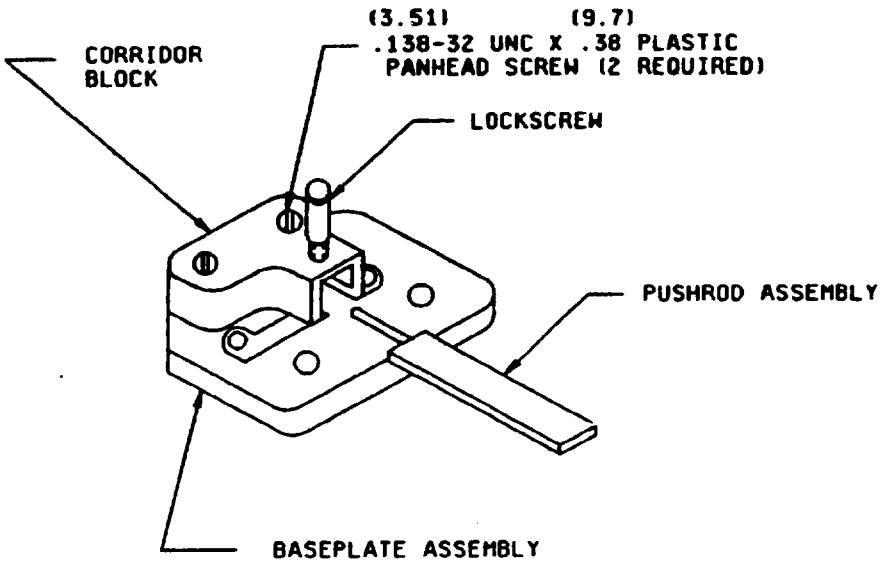


Inches	mm
.0185	0.470
.030	0.76
.037	0.94
.084	2.13
.100	2.54
.125	3.18
.290	7.37
.45	11.4
.800	20.32
1.73	43.94

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. Material: Plastic.

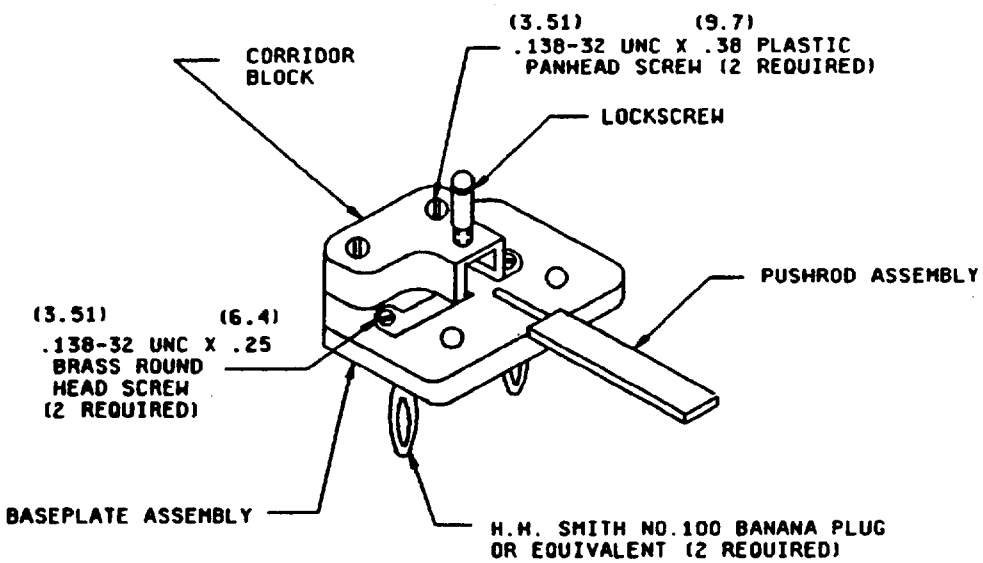
FIGURE 3. Tuning, tool (TT-2271 or equivalent).

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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.

FIGURE 4. Typical chip coil test fixture (TF-250RX-3, or equivalent).



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.

FIGURE 5. Typical chip coil test fixture (TF-2600-3, or equivalent).

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TABLE II. Electrical characteristics (final).

Inspection group	Allowable variation from the initial measurements			
Qualification inspection	Inductance	DC resistance	Self-resonant frequency	Q
	Percent		Percent	Percent
Group II	±5	±(3% +0.001 ohm)	-8	-10
Group IV	±5	±(2% +0.001 ohm)	-10	-10
Group V	±2	- - - - -	- - -	-10
Quality conformance inspection				
Group C				
Subgroup II	±5	±(3% +0.001 ohm)	-8	-10
Subgroup IV	±5	±(3% +0.001 ohm)	-8	-10

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - MI
Navy - AS
Air Force - 19

User activity:
Navy - OS

Preparing activity:
Army - ER

Agent:
DLA - ES

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